ECE 6701 + CS 6501: Optimization for Engineering **Due Fri, Nov 20**

Homework 4: Approximation, Statistics, and Geometry Problems

Instructions: Submit a single Jupyter notebook (.ipynb) of your work to Canvas by 11:59pm on the due date. All math must be formatted using LaTeX. **Be sure to show all the work involved in deriving your answers! If you just give a final answer without explanation, you may not receive credit for that question.**

You may discuss the concepts with your classmates, look on the internet for help, or ask an LLM for advice, but first try hard to solve problems on your own, and always write up the answers entirely on your own. Make sure you understand the concepts and solutions—don't just look up the answers or use AI to answer it for you. **Cite any sources you used outside of the class material (webpages, LLMs, etc.).**

1. Consider the Support Vector Machine primal optimization problem from class. Here we have data $x_i \in \mathbb{R}^d$ with corresponding labels $y_i \in \{-1, +1\}$, for $i = 1, \ldots, N$. The goal is to find weight vector $w \in \mathbb{R}^d$ and bias $b \in \mathbb{R}$. The ζ_i are slack variables.

$$\begin{array}{lll} \text{minimize} & \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{N} \zeta_i & \\ \text{subject to} & y_i(w^T x_i - b) \geq 1 - \zeta_i, & \text{for } i = 1, \ldots, N, \\ & \zeta \succeq 0. & \end{array}$$

   Derive the dual optimization problem.

2. Exercise 7.3 in B&V

3. Exercise 8.7 in B&V

4. **Estimating Gaussian Graphical Models** A Gaussian graphical model is simply a multivariate normal distribution, parameterized by a mean $\mu \in \mathbb{R}^n$ precision matrix $\Omega = \Sigma^{-1} \in S_{++}^n$, the inverse of the covariance matrix. The pdf is

$$p(x; \mu, \Omega) = \frac{1}{(2\pi)^{d/2}|\Omega|^{1/2}} \exp\left(\frac{1}{2}(x-\mu)^T\Omega(x-\mu)\right)$$

   (a) Let $x = (x_1, \ldots, x_n)$ be a multivariate normal random variable as above. Two coordinates x_i and x_j are said to be conditionally independent if

$$p(x_i, x_j \mid \tilde{x}) = p(x_i \mid \tilde{x})p(x_j \mid \tilde{x}),$$

   where $\tilde{x} = \{x_k : k \neq i, k \neq j\}$. Prove that x_i and x_j are conditionally independent if $\Omega_{ij} = 0$.

   (b) Write down a convex optimization problem to estimate a sparse precision matrix Ω of a multivariate normal. Assume that μ is a known constant. Use the multivariate normal log-likelihood function and an L_1 log-prior (a.k.a. Laplace prior) on the off-diagonal entries of Ω, i.e., the Ω_{ij} with $i \neq j$. Include a parameter γ that weights the L_1 term.

(c) Implement your convex optimization problem for a sparse Ω in CVXPY. Synthesize a sample of $N = 1{,}000$ data from a 4-dimensional multivariate normal distribution with a precision matrix

$$\Omega = \begin{pmatrix} 1 & 0.5 & 0 & 0 \\ 0.5 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0.25 \\ 0 & 0 & 0.25 & 1 \end{pmatrix}.$$

Run the optimization in CVXPY and tune the value of γ to find a "sweet spot" where you do a decent job recovering Ω. Show a run where γ is much smaller and run where γ is much larger. What changes in the optimal Ω do you observe?

(d) Using the provided starter code, estimate a sparse precision matrix Ω representing a Gaussian graphical model for the brain network data contained in the notebook. Plot the Ω as a colored(matrix and also as a graph on the brain (follow the code examples for plots). Compare this to plots of the covariance matrix Σ and the standard (non-sparse) precision matrix $\Omega = \Sigma^{-1}$. (Note: you might play with the γ parameter to see different levels of sparsity in the graph.)